

```
#include <stdarg.h>
int findmax ( int, ... );
void main()
{
    int max;

    max = findmax ( 5, 23, 15, 1, 92, 50 );
    printf ( "\nmaximum = %d", max );

    max = findmax ( 3, 100, 300, 29 );
    printf ( "\nmaximum = %d", max );
}

int findmax ( int tot_num, ... )
{
    int max, count, num;

    va_list ptr;

    va_start ( ptr, tot_num );
    max = va_arg ( ptr, int );

    for ( count = 1; count < tot_num; count++ )
    {
        num = va_arg ( ptr, int );
        if ( num > max )
            max = num;
    }

    return ( max );
}
```

Note how the **findmax()** function has been declared. The ellipses (...) indicate that the number of arguments after the first argument would be variable.

Here we are making two calls to **findmax()**, first time to find maximum out of 5 values and second time to find maximum out of

3 values. Note that for each call the first argument is the count of arguments that follow the first argument. The value of the first argument passed to **findmax()** is collected in the variable **tot_num**. **findmax()** begins with a declaration of a pointer **ptr** of the type **va_list**. Observe the next statement carefully:

```
va_start ( ptr, tot_num );
```

This statement sets up **ptr** such that it points to the first variable argument in the list. If we are considering the first call to **findmax()**, **ptr** would now point to 23. The statement **max = va_arg (ptr, int)** would assign the integer being pointed to by **ptr** to **max**. Thus 23 would be assigned to **max**, and **ptr** would now start pointing to the next argument, i.e 15. The rest of the program is fairly straightforward. We just keep picking up successive numbers in the list and keep comparing them with the latest value in **max**, till all the arguments in the list have been scanned. The final value in **max** is then returned to **main()**.

How about another program to fix your ideas? This one calls a function **display()** which is capable of printing any number of arguments of any type.

```
#include <stdio.h>
#include <stdarg.h>
void display ( int, int, ... );
void main()
{
    display ( 1, 2, 5, 6 );
    display ( 2, 4, 'A', 'a', 'b', 'c' );
    display ( 3, 3, 2.5, 299.3, -1.0 );
}

void display ( int type, int num, ... )
{
    int i, j;
    char c;
```